

README for Blue Mountain EGS DAS Dataset

Dataset Description

This dataset was collected as part of Fervo Energy's Blue Mountain Enhanced Geothermal Systems (EGS) demonstration in northern Nevada, near an active geothermal power plant. The project deployed a first-of-its-kind horizontal injector–producer well pair to evaluate reservoir performance in high-temperature, hard-rock conditions.

A 37-day crossflow production test was conducted to characterize reservoir permeability and fracture connectivity. The vertical monitoring well (73-22) was equipped with a permanent fiber optic monitoring cable behind casing, providing real-time DAS data for microseismic monitoring. This study analyzes three key parts of the experiment: (1) the stimulation of the 34-22 production well, (2) a crossflow production test lasting approximately 37 days to establish flow between the producer and injector wells, and (3) a series of dispatchable load cycles, during which injection and/or production were increased. Throughout the experiment, a bottom-hole pressure gauge installed outside the well casing at well 73-22 directly measured reservoir pressure. We use the pressure data to corroborate the observed spatio-temporal trends in microseismicity at the site. The drilling and stimulation phase took place between January 2022 and March 2023, with well testing occurring from April 2023 to May 2023.

The real-time monitoring system developed in this study operated between March 15 and May 23, enabling the tracking of microseismicity related to production stimulation, crossflow tests, and dispatchable loading cycles simulating flexible EGS system operation. All seismicity remained below magnitude 2.0, within the project's Traffic Light System (TLS) Green threshold, with no operational disruptions.

Citation

If you use this dataset, please cite:

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License

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- **Code:** All code in this repository is licensed under the [MIT License](https://opensource.org/licenses/MIT).

Keywords

Distributed Acoustic Sensing, DAS, microseismicity, geothermal, induced seismicity, real-time monitoring, Blue Mountain, EGS, Nevada

FAIR Compliance

This dataset follows the FAIR principles:

- **Findable:** Hosted on OSF with DOI and structured metadata.
- **Accessible:** Publicly available without login barriers.
- **Interoperable:** Provided in open, standard formats (.txt, .mat, .h5).
- **Reusable:** Includes all code and documentation to enable replication and extension.

Technical Specifications

This open dataset includes:

- A real-time DAS-based microseismic event catalog
- Raw and processed continuous DAS examples
- Time series of injection, production, and pressure
- Scripts and input data used for real-time event location processing
- Diffusivity analysis files

The dataset is intended to support reproducibility, benchmarking of DAS-based microseismic workflows, and further research into pressure-induced seismicity and real-time EGS monitoring.

Folder Structure

❖ **BM_DAS_RT_microseismic_catalog_with_locations** / Contains catalog files obtained from real-time analysis

- BM_event_DAS_edgeproc_catalog.txt
- Catalog_Data.mat
- make_catalog.m

❖ **Diffusivity_curves_and_distances_of_events_from_the_well** / Contains data and script to plot diffusivity curve

- Diffusion_data.mat
- Plot_diffusion.m

➤ Vertical_distance_from_injection_FLEX.txt

❖ **Exemplary_Continuous_DAS_recordings_from_the_observation_well_73-22** / Contains exemplary raw DAS recordings

➤ BM73-22.UTC_20230203_052437.149.h5

➤ BM73-22.UTC_20230203_052637.149.h5

➤ BM73-22.UTC_20230203_081837.149.h5

❖ **Injection_Production_and_Pressure_Time_Series** / Contains production, injection and pressure time series

➤ Injection_Production_Rate_Time_Series.txt

➤ Pressure_Time_Series.txt

❖ **Location_code_and_exemplary_application** / Contains data and codes required to replicate the location processing that was performed in real-time during Blue Mountain experiment

➤ DAS_event.mat

➤ RT_DAS_2D_location.m

➤ smooth2a.m

File Manifest

File Name	Format	Description	Relationship
BM_event_DAS_edgeproc_catalog	.txt	Microseismic catalog from this study	Output of make_catalog
Catalog_Data	.mat	Catalog in MATLAB format	Input for make_catalog
make_catalog	.m	MATLAB script to make catalog file	Converts mat catalog to txt
Diffusion_data	.mat	Diffusion time series	Input for plot_diffusion
Plot_diffusion	.m	Script to plot diffusion curve	Uses Diffusion_data
Vertical_distance_from_injection_FLEX.txt	.txt	Vertical event information	Vertical event information used for Figure 6a
BM73-22.UTC_20230203_052437.149	.h5	1-min continuous DAS recording	Exemplary raw continuous DAS data

BM73-22.UTC_20230203_052637.149	.h5	1-min continuous DAS recording	Same as above
BM73-22.UTC_20230203_081837.149	.h5	1-min continuous DAS recording	Same as above
Injection_Production_Rate_Time_Series	.txt	Injection and production rate time series	Supporting data
Pressure_Time_Series	.txt	Bottom-hole pressure time series	Supporting data
DAS_event	.mat	Preprocessed DAS event recording	Input for RT_DAS_2D_location script
RT_DAS_2D_location	.m	Real-time event location estimation script	Locates events using DAS_event
smooth2a	.m	2D Gaussian smoothing script	Post-processes location output